

Identification of the Minimum Effective Dose for Normally Distributed Endpoints Using a Model Selection Approach

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Abstract

When identifying the minimum effective dose (MED) or the lowest observed adverse event level (LOAEL), researchers usually employ multiple comparison procedures (MCPs). The ones preferred are the Dunnett-type difference-to-placebo and ratio-to-control test. In this article, we will use a model selection criterion, namely the generalized order-restricted information criterion (GORIC).

The GORIC can evaluate a set of hypotheses regarding the response means directly and simultaneously, where in this case in each hypotheses a different dose is hypothesized to be the MED or LOAEL. It takes different patterns of increasing response with increasing dose into account without pooling response means, whereas MCPs with order restrictions do. The GORIC chooses the best hypothesis of the set leading to the identification of the MED and, depending on the set, to a specific pattern of response means. We will show, by simulation, that the GORIC has advantages in identifying the MED or LOAEL, besides theoretical ones.

Key Words: Analysis of variance; GORIC; Optimal dose; Order restrictions; LOAEL; MED

1 Introduction

Selecting an appropriate dose for a new drug entity is a challenging problem. A specific aspect is the identification of the minimum effective dose (MED) for a primary efficacy endpoint in a randomized clinical trial phase II [Committee for Proprietary Medicine Products, 2002]. A related problem is the identification of the lowest observed adverse event level (LOAEL) [Kodell, 2009], which is important in toxicological risk assessment. As adverse

event, we consider here a normally distributed endpoint, such as body weight in a repeated toxicity study. Note that the MED and the LOAEL are statistically similar approaches: each determines the lowest dose for which the mean response differs significantly from the placebo or control and for which the consecutive doses have at least the same or increasing differences to the control. The two only differ in application: the first is often employed in efficacy measures and the latter in safety measures/adverse effects. In the sequel, we will refer to both as the MED, except in the application of a safety measure.

A pioneering paper [Ruberg, 1989] proposed different types of contrast tests to estimate the MED. However, Bauer [1997] mentioned the need of the strict monotonicity assumption for an unbiased MED estimate, and Hothorn and Hauschke [2000] demonstrated the possibility of false significant effects in smaller doses when pooling groups. Therefore, only the many-to-one comparison procedure for difference-to-placebo [Dunnett, 1955] or for ratio-to-control [Dilba et al., 2004] reveal unbiased estimates. Nevertheless, the evaluation of dose-finding trials by multiple pooling contrasts was recently recommended [Aras et al., 2011]. Moreover, MED identification by means of statistical testing approaches can be misleading since it depends on the design of the study, particularly on sample size [Hung and Wang, 2010, Leisenring and Ryan, 1992].

In this article, we introduce the model selection approach called the generalized order-restricted information criterion (GORIC) [Kuiper et al., 2011] as an alternative to identify a MED. It evaluates hypotheses of interest, possibly including order restrictions, directly and simultaneously; and is oriented to select the best model. All kinds of models can be evaluated. But, for the identification of the MED, one should examine those where a

specific dose is hypothesized to be the MED. Like other information criteria, it consists of a likelihood and a penalty part. The GORIC is a modification of the model selection approach of Yanagawa and Kikuchi [2001] such that it takes the monotonicity of the dose-response relationship into account (in determining the penalty) and not solely the number of parameters.

There are several differences between an MCP and the GORIC. One, the GORIC is able to take different patterns of increasing response with increasing dose into account (simultaneously), while an MCP rejects a null against an alternative in which two (possibly pooled) means differ. Second, the GORIC examines patterns of response means without the need of pooling means, whereas an MCP with order restrictions requires to pool means. Third, where an MCP leads to a dichotomous decision of a specific dose being a MED or not, the GORIC (weights) can also render relative support. Fourth, an MCP controls the familywise error rate (FWER), that is, the probability of making one or more false discoveries. In contrast, the GORIC/model selection does not (and cannot), since it serves another goal. In model selection, all hypotheses are of equal importance, while the null hypothesis is of utmost importance in hypothesis testing. The latter controls the FWER to ensure that the probability of making one or more false discoveries does not exceed the nominal alpha level (often set to .05). In model selection, one evaluates hypotheses directly and chooses the best of a set, thus in that sense there is no need for FWER control. Notably, in model selection, the equivalent of the Type-I error (i.e., incorrectly rejecting a true null hypothesis) does exist, namely not selecting the true null hypothesis (when it is included in the set). This is also not controlled for, since the null hypothesis does not serve a special role here.

To facilitate the description of the two methods, we will illustrate them by means of two data examples, which are introduced in Section 2. A brief description of the Dunnett-type difference-to-placebo test, in the sequel referred to as the Dunnett test, and the GORIC will be given in Section 3 for identifying the MED. Subsequently, in Section 4, these methods will be applied to the two data examples in which the MED is searched for in the first and, due to the type of application, the LOAEL is obtained in the second. In Section 5, the Dunnett test, accompanied with the Williams test for order-restricted comparison against control [Williams, 1971], and the GORIC will be examined by means of simulation. Here, it is shown that the Williams procedure is biased (as other to-the-left pooling contrasts). Therefore, this method is not elaborated upon in the Methods and Examples sections (i.e., Sections 3 and 4, respectively). We end with alternatives for LOAEL and some extensions of (and software for) the GORIC in Section 6 and with a discussion in Section 7.

2 Motivating examples

The first data example is part of a dose response clinical trial regarding a drug to treat angina pectoris. The primary efficacy endpoint is the change in pain-free walking from pre-treatment (measured in minutes), that is, the duration of pain-free walking after treatment relative to the values before treatment. The data were taken from Westfall et al. [1999, p. 164] and are available under the name “angina” in the R package `mratios`. Figure 1 displays the boxplots of change in pain-free walking for four dose groups and zero-dose placebo. Bear in mind that large values indicate positive effects on patients.

As a second data example, a chronic toxicity study on Mosapride Citrate [Fitzhugh

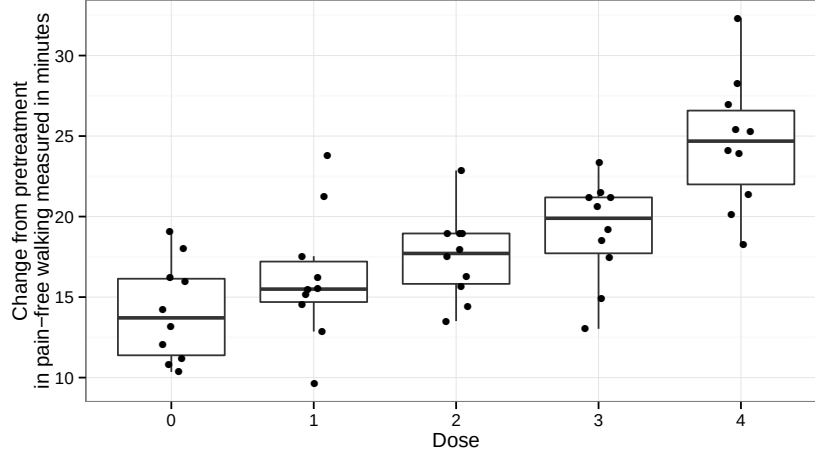


Figure 1: Boxplots of change in pain-free walking (measured in minutes) for five drug doses to treat angina pectoris

et al., 1964] was selected, regarding the relative liver weight (relative to the body weight) of dogs. Due to the application, one searches for the LOAEL here (instead of the MED). These data were employed by Yanagawa and Kikuchi [2001] and West and Kodell [2005] and are available from Table 2 on page 320 of Yanagawa and Kikuchi [2001]. Figure 2 displays the boxplots of relative liver weight of dogs for three Mosapride Citrate levels and the zero-dose control.

3 Methods

3.1 Definition of MED

Consider one-way layouts with the factor *dose* with $i = 0, 1, 2, \dots, k$ levels, where the placebo group or the negative control group is coded as a zero-dose control (i.e., $i = 0$). For instance, in the first example, $k = 4$ and, in the second, $k = 3$. Furthermore, let Y_{ij}

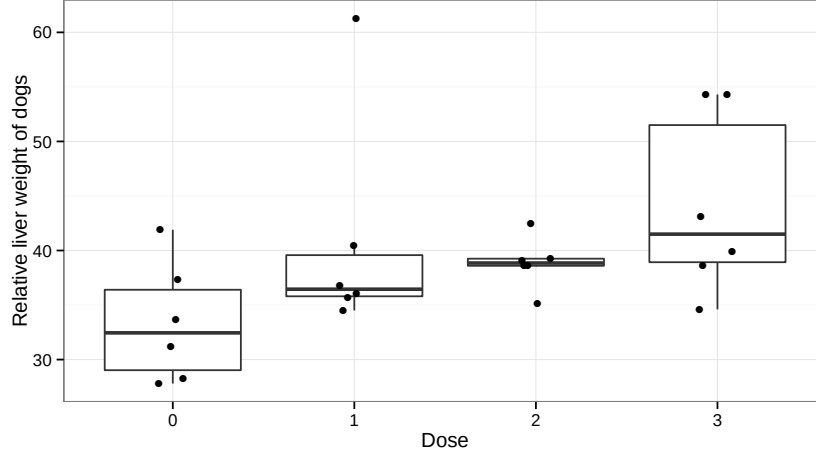


Figure 2: Boxplots of relative liver weight of dogs for four doses of Mosapride Citrate

be a primary (efficacy or safety) endpoint which is normally distributed with mean μ_i for group i and a homogeneous variance σ^2 , that is, $Y_{ij} \sim N(\mu_i, \sigma^2)$. The definition of the MED (or LOAEL) is:

$$MED = \min(\eta \in 1, \dots, k: \mu_0 = \dots = \mu_{\eta-1} < \mu_\eta \leq \dots \leq \mu_k).$$

3.2 Contrast tests

In a testing framework, the global null hypothesis

$$H_0 : \quad \mu_0 = \mu_1 = \dots = \mu_k$$

can be compared to a global order-restricted alternative

$$H_A^{ordered} : \quad \mu_0 = \mu_1 = \dots = \mu_{\eta-1} < \mu_\eta \leq \dots \leq \mu_k,$$

where η is the unknown MED. For instance, for $k = 3$, the global alternative can be decomposed in seven elementary order-restricted alternatives [Bretz, 1999]:

$$\begin{aligned}
H_{\eta=1}^a : \quad & \mu_0 < \mu_1 = \mu_2 = \mu_3 \\
H_{\eta=1}^b : \quad & \mu_0 < \mu_1 < \mu_2 = \mu_3 \\
H_{\eta=1}^c : \quad & \mu_0 < \mu_1 = \mu_2 < \mu_3 \\
H_{\eta=1}^d : \quad & \mu_0 < \mu_1 < \mu_2 < \mu_3 \\
H_{\eta=2}^a : \quad & \mu_0 = \mu_1 < \mu_2 = \mu_3 \\
H_{\eta=2}^b : \quad & \mu_0 = \mu_1 < \mu_2 < \mu_3 \\
H_{\eta=3}^a : \quad & \mu_0 = \mu_1 = \mu_2 < \mu_3,
\end{aligned} \tag{1}$$

where $H_{\eta=e}^x$ denotes a specific pattern of response means, here labeled as x , when dose e is the MED. A subset of these alternatives, namely $H_{\eta=1}^a$, $H_{\eta=2}^a$, and $H_{\eta=3}^a$, are change point alternatives [Hirotzu et al., 2011], in which some response means are pooled in a testing framework (namely, doses 1, 2, & 3 in $H_{\eta=1}^a$ and doses 2 & 3 in $H_{\eta=2}^a$). As mentioned before, all testing approaches which use pooling means (e.g., contrasts) render biased MED estimations. For example, $H_{\eta=1}^a: \mu_0 < \mu_1 = \mu_2 = \mu_3$ can be false, while $\mu_0 < \bar{\mu}_{123}$ is true, with $\bar{\mu}_{123}$ the pooled mean (i.e., the (weighted) average) of μ_1 , μ_2 , and μ_3 . Therefore, these tests prefer low(er) doses as the MED and, thus, they can underestimate the MED.

3.3 The Dunnett test

The Dunnett test [Dunnett, 1955] is an unbiased testing approach for the identification of the MED, which employs linear combinations of coefficients (here, means) of the $k + 1$ dose

groups. In the first example, where $k = 4$, the alternatives examined with the one-sided many-to-one comparisons of Dunnett are:

$$H_1 : \quad \mu_0 < \mu_1, \mu_2, \mu_3, \mu_4$$

$$H_2 : \quad \mu_0 < \mu_2, \mu_1, \mu_3, \mu_4$$

$$H_3 : \quad \mu_0 < \mu_3, \mu_1, \mu_2, \mu_4$$

$$H_4 : \quad \mu_0 < \mu_4, \mu_1, \mu_2, \mu_3.$$

As a result, the following hypotheses are tested simultaneously with a one-sided Dunnett test:

$$\mu_1 - \mu_0 \leq 0 \quad \text{versus} \quad \mu_1 - \mu_0 > 0$$

$$\mu_2 - \mu_0 \leq 0 \quad \text{versus} \quad \mu_2 - \mu_0 > 0$$

$$\mu_3 - \mu_0 \leq 0 \quad \text{versus} \quad \mu_3 - \mu_0 > 0$$

$$\mu_4 - \mu_0 \leq 0 \quad \text{versus} \quad \mu_4 - \mu_0 > 0$$

This procedure is available in the R package `multcomp` [Bretz et al., 2002], which renders so-called simultaneous p -values. The lowest dose for which the one-sided simultaneously p -value (adjusted for multiplicity, that is, controlling the FWER) is just below the nominal alpha level (often, and also here, set to .05) is the MED. For the second example, the hypotheses tested to identify the LOAEL are equal to those used in the first example with exception of the last hypothesis, since now $k = 3$.

3.4 Model selection for order-restricted hypotheses

In contrast to testing various null hypotheses versus an alternative, model selection evaluates a set of hypotheses of interest directly and simultaneously. One such model selection technique is the generalized order-restricted information criterion [GORIC; Kuiper et al., 2011, 2012]. The GORIC is, like the AIC [Akaike, 1973] and ORIC [Anraku, 1999], a trade-off between the fit of the hypothesis in the data (which is related to the likelihood) and the complexity of the hypothesis (which is related to the number of distinct parameter values). The calculation of the GORIC is given in Appendix A.

The GORIC selects the best out of a set of hypotheses/models. One can best construct this set based on theory/expectations. For instance, when $k = 3$, one could examine the seven elementary order-restricted alternatives specified in (1) if one is interested in which of those is the correct ordering. If one is just interested in the change point alternatives $H_{\eta=1}^a$, $H_{\eta=2}^a$, and $H_{\eta=3}^a$, then one should inspect only those (notably, without the necessity of pooling means). The hypothesis with the lowest GORIC value is the preferred one (of the set).

To protect against selecting the best out of a set of weak hypotheses (i.e., hypotheses not strongly supported by the data), one should always incorporate the unconstrained hypothesis (i.e., the hypothesis with no restrictions on the parameters, also known as the classical or omnibus alternative hypothesis). Then, the GORIC selects (i) the correct hypothesis (if it is included), (ii) a similar one (i.e., a hypothesis which resembles the true hypothesis, that is, only differs in a few constraints), or (iii) the unconstrained hypothesis. Note that, if the set does not contain the null model and the number of observations is

high enough, the unconstrained model will safeguarded for preferring a non-effective dose. One could also include the null in the set to protect against selecting a non-effective dose.

To improve interpretation, we will report the GORIC weights (w_m), which are elaborated upon in Appendix B. These weights reflect the relative support of one hypothesis of interest in comparison to the whole set. Moreover, the ratio of two weights gives the relative support of these two hypotheses: H_m is $w_m/w_{m'}$ times more likely than $H_{m'}$.

3.4.1 Hypotheses of interest

We believe that there are often two types of research questions: One wants to identify the MED level or one wants additionally to obtain information about the pattern of the response means, which can be useful in clinical trials. In the latter case, one should evaluate all hypotheses based on the decomposition of the global order-restricted alternative $H_A^{ordered}$: $\mu_0 \leq \mu_1 \leq \dots \leq \mu_k$, with at least one strict inequality constraint, and the unconstrained hypothesis H_u . This will lead to a preferred pattern/shape of response means for a specific MED level. For instance, in (1), $H_{\eta=1}^a$ to $H_{\eta=1}^d$ represent patterns where dose 1 is the MED and model selection might lead to preferring one of those four patterns. In the first case, when one is not interested in the specific pattern of the response means, but solely in the identification of the MED, one should examine only those with a monotonic increase of means after μ_η (e.g., $H_{\eta=1}^d$, $H_{\eta=2}^b$, and $H_{\eta=3}^a$ in (1)).

For those who are mathematically inclined, it holds true for the GORIC that “ $<$ ” equals “ $\leq$ ”. Hence, in (1), $H_{\eta=1}^a$ is a subset/special case of $H_{\eta=1}^d$. Therefore, when $H_{\eta=1}^a$ is true, $H_{\eta=1}^d$ will be true as well; this is reflected by the same likelihood value. This also holds true for the other patterns (here: $H_{\eta=1}^b$ and $H_{\eta=1}^c$). Thus, the hypotheses with a monotonic

increase of means after μ_η (e.g., $H_{\eta=1}^d$ in (1)) covers all patterns of η ($= 1$). Notably, when both $H_{\eta=1}^a$ and $H_{\eta=1}^d$ are true, the hypotheses are distinguished by their complexity; where here pattern a has a lower penalty value than d and is thus preferred. This is, besides a theoretical one, the reason why one should examine the patterns separately when it is of interest. Namely, it might be possible that pattern $H_{\eta=1}^a$ is the correct one, but that the support for $H_{\eta=1}^d$ is not convincing (compared to, say, $H_{\eta=2}^b$).

In both examples, we will first examine all hypotheses based on the decomposition of the global order-restricted alternative and the unconstrained hypothesis (H_u). For brevity, the sets of models are not displayed here, but can be found in Table 2 for the first example and in Table 4 for the second one. Subsequently, we will investigate solely the hypotheses with a monotonic increase of means after μ_η . For the first example, the set then comprises $H_{\eta=1}^h$, $H_{\eta=2}^d$, $H_{\eta=3}^b$, $H_{\eta=4}^a$, and H_u of Table 2. Similarly, $H_{\eta=1}^d$, $H_{\eta=2}^b$, $H_{\eta=3}^a$, and H_u of Table 4 will be examined in the second example.

To make sure that two means differ with at least a specific amount, say δ , one could for the second example, where $k = 3$, evaluate:

$$\begin{aligned}
H_0 : \quad & \mu_0 = \mu_1 = \mu_2 = \mu_3 \\
H_{\eta=1} : \quad & \mu_0 + \delta \leq \mu_1 \leq \mu_2 \leq \mu_3 \\
H_{\eta=2} : \quad & \mu_0 = \mu_1, \mu_1 + \delta \leq \mu_2 \leq \mu_3 \\
H_{\eta=3} : \quad & \mu_0 = \mu_1 = \mu_2, \mu_2 + \delta \leq \mu_3 \\
H_u : \quad & \mu_0, \mu_1, \mu_2, \mu_3.
\end{aligned} \tag{2}$$

This type of set is not employed in the examples, but will be used in the simulation study (except for H_u). Notably, when $\delta = 0$, this set reduces to H_0 , $H_{\eta=1}^d$, $H_{\eta=2}^b$, $H_{\eta=3}^a$, and H_u .

of Table 4.

4 Evaluation of the examples

4.1 Example 1: Change in pain-free walking

Performing the one-sided Dunnett test renders so-called one-sided simultaneous p -values. Table 1 shows that for $\alpha = .05$, when taking multiplicity into account (hence, based on adjusted p -values), dose $\eta = 3$ is the MED.

Table 1: *One-sided simultaneous p -values (adjusted for multiplicity) of the Dunnett test regarding change in pain-free walking*

Null hypothesis	Estimate	Adjusted p -value
$\mu_1 - \mu_0 \leq 0$	2.10	.239
$\mu_2 - \mu_0 \leq 0$	3.40	.053
$\mu_3 - \mu_0 \leq 0$	5.00	.004
$\mu_4 - \mu_0 \leq 0$	10.50	< .001

Table 2 (for now, disregard the last column) shows, for each of the elementary order-restricted alternatives and H_u , the values for the order-restricted log likelihood, the penalty, the GORIC, and the GORIC weights regarding change in pain-free walking. Since $H_{\eta=1}^h: \mu_0 < \mu_1 < \mu_2 < \mu_3 < \mu_4$ has the lowest GORIC value and, therefore, the highest GORIC weight, it is the preferred hypothesis of the set. Moreover, $H_{\eta=1}^h$ is about $.275/.018 \approx 15.28$ fold more likely than the unconstrained hypothesis. Hence, $H_{\eta=1}^h$ is not a weak hypothesis. Thus, according to the GORIC, based on these data and this set of hypotheses, dose $\eta = 1$

is the MED, and the best ordering of the response means is $\mu_0 < \mu_1 < \mu_2 < \mu_3 < \mu_4$. However, it has only $.275/.130 \approx 2.12$ more support than $H_{\eta=2}^d: \mu_0 = \mu_1 < \mu_2 < \mu_3 < \mu_4$, the best hypothesis for $\eta = 2$. One could increase confidence in rejecting this pattern as the preferred one by investigating a larger data set. Remarkably, $H_{\eta=1}^h$ is $.275/.034 \approx 8.09$ more likely than the best hypothesis for $\eta = 3$, the MED chosen by the one-sided Dunnett test.

The GORIC weights for the hypotheses in which all patterns leading to a specific dose are combined are reported in the last column of Table 2. Also now, $H_{\eta=1}^h$ is the preferred hypothesis of the set and, thus, dose $\eta = 1$ is the MED. Since the hypotheses in this smaller set are the ones with the highest support for each η in the whole set, there are practically no differences in relative weights. That is, $H_{\eta=1}^h$ is here about $.598/.038 \approx 15.18$ times more likely than the unconstrained hypothesis (and is, therefore, not a weak hypothesis). Furthermore, it has only $.598/.282 \approx 2.12$ more support than $H_{\eta=2}^d$, where the MED is 2. One could increase confidence in dismissing dose $\eta = 2$ as the MED by inspecting a larger data set. Moreover, $\eta = 1$ (i.e., $H_{\eta=1}^h$) is $.598/.075 \approx 7.97$ more likely than $\eta = 3$ (i.e., $H_{\eta=3}^b$), the MED chosen by the one-sided Dunnett test.

Table 2: *GORIC values regarding change in pain-free walking*

Model	ORLL	Penalty	GORIC	(MED + pattern)	GORIC	GORIC
					weights	weights
					(MED)	(MED)
$H_{\eta=1}^a: \mu_0 < \mu_1 = \mu_2 = \mu_3 = \mu_4$	-144.55	2.50	294.11		.000	
$H_{\eta=1}^b: \mu_0 < \mu_1 < \mu_2 = \mu_3 = \mu_4$	-140.80	2.78	287.16		.000	
$H_{\eta=1}^c: \mu_0 < \mu_1 = \mu_2 < \mu_3 = \mu_4$	-136.92	2.87	279.57		.001	
$H_{\eta=1}^d: \mu_0 < \mu_1 = \mu_2 = \mu_3 < \mu_4$	-132.31	2.91	270.44		.060	
$H_{\eta=1}^e: \mu_0 < \mu_1 < \mu_2 < \mu_3 = \mu_4$	-136.61	3.03	279.29		.001	
$H_{\eta=1}^f: \mu_0 < \mu_1 < \mu_2 = \mu_3 < \mu_4$	-131.01	3.13	268.28		.178	
$H_{\eta=1}^g: \mu_0 < \mu_1 = \mu_2 < \mu_3 < \mu_4$	-130.82	3.14	267.92		.214	
$H_{\eta=1}^h: \mu_0 < \mu_1 < \mu_2 < \mu_3 < \mu_4$	-130.43	3.28	267.42		.275	.598
$H_{\eta=2}^a: \mu_0 = \mu_1 < \mu_2 = \mu_3 = \mu_4$	-141.46	2.50	287.92		.000	
$H_{\eta=2}^b: \mu_0 = \mu_1 < \mu_2 < \mu_3 = \mu_4$	-137.39	2.77	280.33		.000	
$H_{\eta=2}^c: \mu_0 = \mu_1 < \mu_2 = \mu_3 < \mu_4$	-131.99	2.87	269.72		.087	
$H_{\eta=2}^d: \mu_0 = \mu_1 < \mu_2 < \mu_3 < \mu_4$	-131.42	3.04	268.92		.130	.282
$H_{\eta=3}^a: \mu_0 = \mu_1 = \mu_2 < \mu_3 = \mu_4$	-138.65	2.50	282.30		.000	
$H_{\eta=3}^b: \mu_0 = \mu_1 = \mu_2 < \mu_3 < \mu_4$	-133.01	2.78	271.57		.034	.075
$H_{\eta=4}^a: \mu_0 = \mu_1 = \mu_2 = \mu_3 < \mu_4$	-135.97	2.51	276.94		.002	.005
$H_u: \mu_0, \mu_1, \mu_2, \mu_3, \mu_4$	-130.43	6.00	272.86		.018	.039

Note. ORLL = order-restricted log likelihood; GORIC = generalized order-restricted information criterion = 2 (ORLL - penalty).

4.2 Example 2: Relative liver weight

Table 3 depicts the one-sided simultaneous p -values (adjusted for multiplicity) obtained by the one-sided Dunnett test for the second example. This table shows that, for $\alpha = .05$, dose $\eta = 3$ is the LOAEL in the chronic toxicity study regarding the relative liver weight (relative to body weight) of dogs.

Table 3: *One-sided simultaneous p -values (adjusted for multiplicity) of the Dunnett test regarding the relative liver weight of dogs*

Null hypothesis	Estimate	Adjusted p -value
$\mu_1 - \mu_0 \leq 0$	7.450	.105
$\mu_2 - \mu_0 \leq 0$	5.500	.219
$\mu_3 - \mu_0 \leq 0$	10.767	.023

Table 4 (for now, disregard the last column) shows, for each of the seven elementary order-restricted alternatives and H_u , the values for the order-restricted log likelihood, the penalty, the GORIC, and the GORIC weights regarding the relative liver weight of dogs. Since $H_{\eta=1}^c: \mu_0 < \mu_1 = \mu_2 < \mu_3$ has the lowest GORIC value and, therefore, the highest GORIC weight, it is the preferred hypothesis of the set. Moreover, $H_{\eta=1}^c$ is about $.267/.037 \approx 7.22$ times more likely than the unconstrained (i.e., any ordering). Hence, it is not a weak hypothesis. Consequently, according to the GORIC, dose $\eta = 1$ is the LOAEL and $\mu_0 < \mu_1 = \mu_2 < \mu_3$ is the preferred ordering of the response means. Notably, all hypotheses with $\eta = 1$ have at least $.133/.068 \approx 1.96$ times more support than those with $\eta = 2$ or 3; and $H_{\eta=1}^c$ itself is at least $.267/.068 \approx 3.93$ fold more likely than any hypothesis with $\eta = 2$ or 3.

Table 4: *GORIC values regarding the relative liver weight of dogs*

Model	ORLL	Penalty	GORIC	(MED + pattern)	GORIC	GORIC
					weights	weights
					(MED)	(MED)
$H_{\eta=1}^a: \mu_0 < \mu_1 = \mu_2 = \mu_3$	-80.32	2.50	165.68		.176	
$H_{\eta=1}^b: \mu_0 < \mu_1 < \mu_2 = \mu_3$	-80.30	2.80	166.20		.133	
$H_{\eta=1}^c: \mu_0 < \mu_1 = \mu_2 < \mu_3$	-79.51	2.89	164.80		.267	
$H_{\eta=1}^d: \mu_0 < \mu_1 < \mu_2 < \mu_3$	-79.51	3.04	165.11		.229	.587
$H_{\eta=2}^a: \mu_0 = \mu_1 < \mu_2 = \mu_3$	-81.95	2.50	168.90		.034	
$H_{\eta=2}^b: \mu_0 = \mu_1 < \mu_2 < \mu_3$	-81.15	2.81	167.92		.056	.144
$H_{\eta=3}^a: \mu_0 = \mu_1 = \mu_2 < \mu_3$	-81.28	2.49	167.53		.068	.174
$H_u: \mu_0, \mu_1, \mu_2, \mu_3, \mu_4$	-79.38	5.00	168.77		.037	.094

Note. ORLL = order-restricted log likelihood; GORIC = generalized order-restricted information criterion = 2 (ORLL - penalty).

Alternatively, one could also inspect the set consisting of just $H_{\eta=1}^d$, $H_{\eta=2}^b$, and $H_{\eta=3}^a$ (where all patterns leading to dose η are combined), accompanied with H_u . The GORIC weights for this set are reported in the last column of Table 4. Now, $H_{\eta=1}^d$ is the favored hypothesis of the set. Like in the set comprising all alternatives, dose $\eta = 1$ is the LOAEL. Furthermore, $H_{\eta=1}^d$ is not a weak hypothesis, since it is about $.587/.094 \approx 6.24$ times more likely than the unconstrained hypothesis. Moreover, $H_{\eta=1}^d$ (i.e., LOAEL of 1) has here about $.587/.144 \approx 4.08$ and $.587/.174 \approx 3.37$ times more support than $H_{\eta=2}^b$ (i.e., LOAEL of 2) and $H_{\eta=3}^a$ (i.e., LOAEL of 3, which was chosen by the one-sided Dunnett

test), respectively.

5 Simulation study

The two examples show that the results of the Dunnett test differ from that of the GORIC. Since it is not known in these examples what the true MED (or LOAEL) is, a simulation study will be performed to obtain more insight in which method performs best (under which circumstances).

The performance of the GORIC is compared with that of the Dunnett- and Williams-type multiple contrast test by simulating the MED selection frequency based on a simple, artificial dose-response experiment with ten observations at each of four increasing, equidistant dose levels. A convex, linear, semi-concave, and concave dose-response profile is assumed, similar to the power calculation examples shown in Genz and Bretz [1999], to cover a wide spectrum of dose-response settings. To study the sensitivity of detecting the true model for each method, the transition from the null model with equal means at each dose level to a clearly detectable dose-response shape is implemented by increasing a non-centrality parameter, as described in Genz and Bretz [1999]. In addition, we scaled this non-centrality parameter to obtain the same effect sizes [Cohen, 1992] for each dose-response profile.

The set of linear constraints in (2), except for H_u , will be examined by the GORIC. To maintain at least a related set of hypotheses to compare the GORIC with multiple testing procedures, H_0 is included in the set of hypotheses and H_u is not. The four models, H_0 to $H_{\eta=3}$, correspond to a MED at dose level 0 to 3, respectively. The parameter δ denotes a

relevance margin, which increases the distance of $H_{\eta=1}$, $H_{\eta=2}$, and $H_{\eta=3}$ to H_0 . The same relevance criterion is introduced to the right hand side of the hypothesis definition of the multiple contrast tests to maintain a rough comparability of the three approaches.

For each of 10,000 simulation runs per setting, the frequency of selecting a specific MED is computed and accordingly the selection rate of the true MED is calculated. The selection rates for the linear true dose-response profile is shown in Figure 3. The results for the convex, semi-concave, and concave dose-response shapes are given in the supplementary material.

With an increasing non-centrality parameter, both the GORIC and the Dunnett-type MCP are able to detect the true underlying MED. In contrast, the Williams-type MCP has difficulties to detect the MED at a semi-concave dose-response shape, as there is no specific hypothesis/contrast available for this case. This can also be seen from the corresponding figure. Note that the four figures show that the Williams-type MCP indeed prefers low(er) doses as the MED, which can lead to the underestimation of the MED in some populations.

The GORIC and MCP approaches can be clearly distinguished by the rate of detecting the null model at a non-centrality parameter of zero. The MCP procedures are defined to select the true null model with a type-I-error of α , whereas the model selection procedure does not give a special weight to the null model, but treats it like any other model in the set. Furthermore, when the non-null model is true, the GORIC often has a higher true-hypothesis rate than the other two methods. It is also less dependent on the non-centrality parameter than the other two and almost independent of it after a certain value (especially for the convex, semi-concave and concave dose-response shape). This suggests favoring the GORIC when one does not expect the null model to be true on forehand. However,

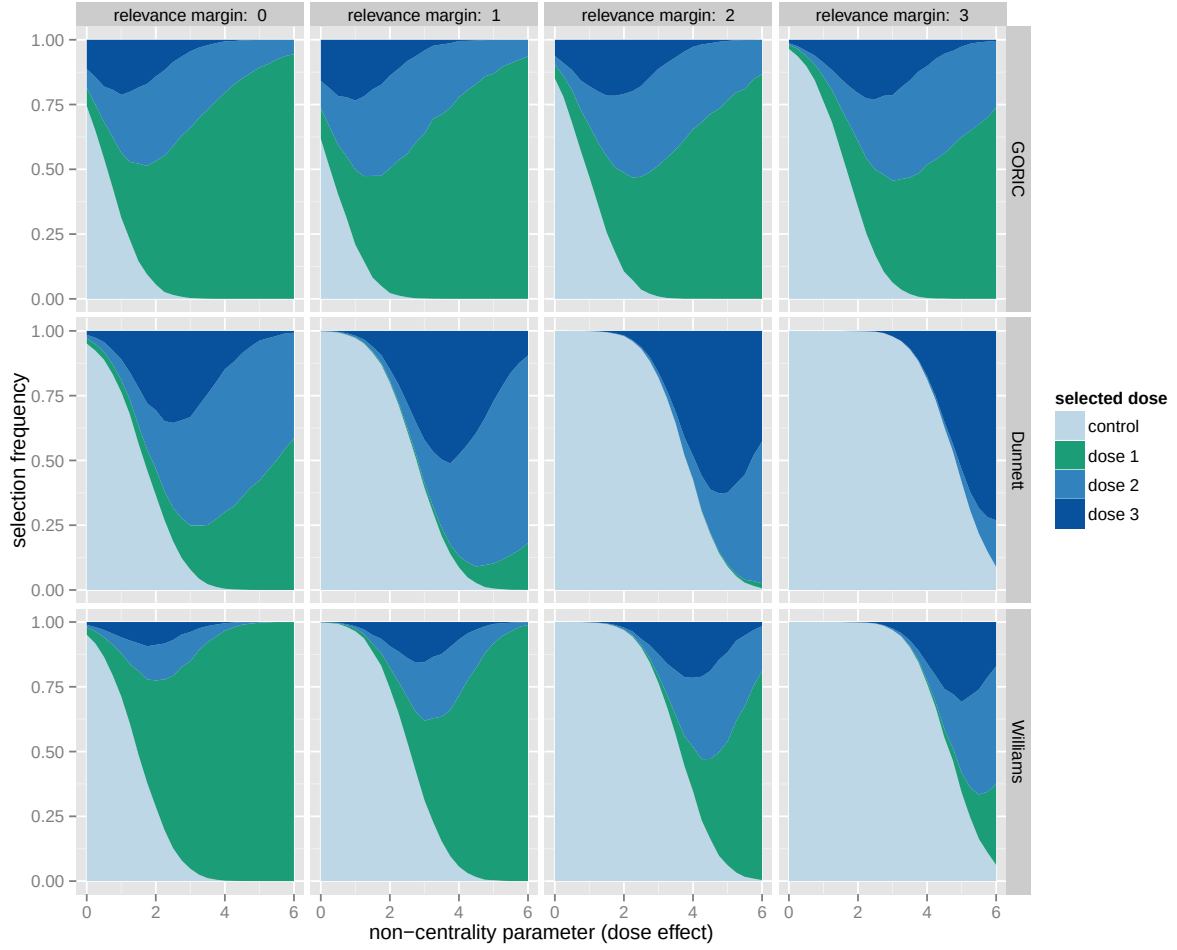


Figure 3: Simulated MED selection rates for the GORIC and the Dunnett- and Williams-type multiple contrast test ($\alpha = .05$) given a non-centrality parameter based on a linear dose-response profile. Several relevance margins ($\delta = 0, 1, 2, 3$) are presented in the columns.

bear in mind that the three methods do not evaluate the same (set of) hypotheses. Thus, besides these properties, one should base the comparison on other characteristics of the three methods. One distinction between the methods is the hypothesis of interest. With MCPs one rejects a null of two means being equal or not (a dichotomous decision), which leads to the MED. Note that often pooled means are required, which can lead to false significant effects in smaller doses [Hothorn and Hauschke, 2000]. The GORIC evaluates a set of theories/hypotheses/ordering of all the means directly and simultaneously. Thus, it can examine very specific orderings of means, which can also lead to the identification of the MED (and might give extra information regarding the pattern of the response means). Here, no pooled means are needed. In addition, via the GORIC weights, one obtains relative support for each pattern of means and/or the value of the MED. Another difference is that MCPs control for the Type-I error or FWER, where the GORIC does not. But, since the GORIC evaluates orderings of means directly and simultaneously, there is no need for controlling the FWER. However, when ‘finding’ the null, when it is true, is of most importance, one should employ an MCP. When the importance lies more in finding a non-null hypothesis, the GORIC is preferred.

6 Extensions

6.1 Alternatives for LOAEL in toxicological risk assessment

According to Hothorn and Hasler [2008], the no-observable-adverse-effect level (NOAEL), controlling the more relevant false-negative error rate, is of more interest than the LOAEL. Additionally, the benchmark dose approach, especially the benchmark dose lower confi-

dence limit approach, is preferred over the LOAEL identification approach [Kodell, 2009]. However, the definition of an appropriate benchmark response level is not clear, see the recent proposed background noise approach based on the difference between the upper and lower bounds of the two-sided 90% confidence interval curves [Sand et al., 2011]. Because of the lower influence of the design, particularly of the non-centrality parameter and thus also of the sample sizes, the identification of the LOAEL by means of the GORIC instead of an MCP is now an alternative to the benchmark dose concept in toxicological risk assessment. Although, it should be stressed that, up to now, the GORIC is restricted to normally distributed endpoints.

6.2 Extensions of the GORIC

There is a version of the GORIC applicable to heterogenous variances [Kuiper, 2011, Kuiper et al., 2012]. For ANOVA models, the adjustment to the GORIC is straightforward. Although, there is not really a biological reasoning behind assuming different variances per dose in the described examples, Table 5 displays the results of the GORIC apt for heterogenous variances for the second example. In this case, $H_{\eta=1}^c: \mu_0 < \mu_1 = \mu_2 < \mu_3$ is still the preferred hypotheses/ordering of means, where the MED is dose 1. Hence, for this example, the GORIC for both homogeneous and heterogenous variances render the same conclusion.

The small sample size of $n_i = 6$, in the second example, is particularly limiting for testing approaches. Fortunately, there exist a small-sample corrected GORIC [Kuiper, 2011, Unpublished], which especially performs better in regression models, when the total number of observations (N) are small (it is shown for $N = 10$ and 20). Notably, in

Table 5: *GORIC values regarding the relative liver weight of dogs - with heterogenous variances*

Model	ORLL	Penalty	GORIC	GORIC
				weights
$H_{\eta=1}^a: \mu_0 < \mu_1 = \mu_2 = \mu_3$	-75.22	2.50	161.44	.140
$H_{\eta=1}^b: \mu_0 < \mu_1 < \mu_2 = \mu_3$	-75.22	2.67	161.78	.118
$H_{\eta=1}^c: \mu_0 < \mu_1 = \mu_2 < \mu_3$	-74.01	2.97	159.96	.294
$H_{\eta=1}^d: \mu_0 < \mu_1 < \mu_2 < \mu_3$	-74.01	3.13	160.28	.251
$H_{\eta=2}^a: \mu_0 = \mu_1 < \mu_2 = \mu_3$	-76.42	2.50	163.83	.042
$H_{\eta=2}^b: \mu_0 = \mu_1 < \mu_2 < \mu_3$	-75.29	2.96	162.49	.083
$H_{\eta=3}^a: \mu_0 = \mu_1 = \mu_2 < \mu_3$	-76.82	2.50	164.64	.028
$H_u: \mu_0, \mu_1, \mu_2, \mu_3, \mu_4$	-73.89	5.00	163.78	.044

Note. ORLL = order-restricted log likelihood; GORIC = generalized order-restricted information criterion = 2 (ORLL - penalty).

ANOVA models, the total number of observations equals the sum of the group sizes (i.e., $N = \sum_{i=0}^k n_i$). In the example, $N = 4 \times 6 = 24$; therefore, the results of the GORIC resemble those of the small-sample corrected GORIC here.

6.3 Software

The GORIC is available in the R package `goric`. There, it can be applied to ANOVA models. There also exist a stand-alone, free-to-use software application of the GORIC for (multivariate) regression models and, thus, ANOVA models. This is downloadable from

<http://www.uu.nl/staff/RMKuiper>.

7 Discussion

Both multiple comparison procedures (MCPs) and the generalized order-restricted information criterion (GORIC) can be used for the identification of the minimum effective dose (MED) or the lowest observed adverse event level (LOAEL), but are based on different principles with substantial different properties. It should be stressed first that the GORIC and MCP serve other goals. Namely, the GORIC selects the best out of a set of hypotheses, whereas MCP rejects the null hypothesis (of two means being equal) or not (in favor of an alternative). Since the GORIC examines the ordering of all response means, there is no need to pool response means or to hypothesize only the relation between two means instead of all of them. Furthermore, MCPs control the familywise error rate (FWER), while the GORIC controls neither false-negative nor false-positive error rate at all, since the null hypothesis has no special interest. Shang [2010] states about the posterior probabilities of the Schwarz information criterion (SIC), which are comparable to the GORIC weights, that “due to the simplicity of the utilized priors for the SIC variant, no adjustments in inferences are needed in conducting multiple comparisons”. Besides that, we believe that no FWER control is required, because the ordering of the means representing a specific MED value are investigated directly and simultaneously. Since the null hypothesis plays no pivotal role in model selection, there is also no (need for a) Type-I error control.

From the examples, it can be seen that by including hypotheses to the set, the weights for a specific pattern of means will not increase and generally decrease. Thus, only evaluate

all alternatives when you are interested in the patterns (as well). However, at least in these examples, (i) the relative weights are not influenced considerably and (ii) the support regarding a specific MED (including all possible patterns) remains about the same in both sets. In that sense, there seems to be no need for adjusting for the number of hypotheses in the set.

Simulation shows, among others, that the GORIC often has a higher true-hypothesis rate, when a non-null hypothesis is true, and that it is particularly less dependent on the non-centrality parameter and thus on the sample sizes. But, bear in mind that the methods do not evaluate the same set of hypotheses. Nevertheless, the GORIC also has theoretical advantages, like evaluating a possible MED directly in one hypothesis and not requiring pooled means. However, when a researcher expects the null model to be true on forehand or pays special importance to the null, it is better not to use the GORIC, since MCPs are designed to choose the null model frequently when it is true. Since we believe that the null will usually not be the true hypothesis (and due to the other advantages), we prefer the use of the GORIC.

To control the Type-I error, one could employ a two-step approach as well. In the first step, evidence for an overall effect of an increase in dose levels on the endpoint is identified by a hypothesis test (at a specific Type-I error rate). If a significant effect is found, the GORIC can be used in a second step to identify the MED.

From the examples, it can be seen that different types of sets can be evaluated with the GORIC. Which set should be employed in practice depends on the research question. The use of the set where one hypothesis contains all possible orderings representing a specific dose to be the MED is sufficient for identifying the MED. When one can specify a

minimum difference or wants to evaluate a specific minimum difference (say, δ), one should use the set used in the simulation study accompanied with the unconstrained model, that is, the set displayed in (2). In case one is interested in which of all the possible patterns (resulting from the decomposition of the global order-restricted alternative) is the best theory (besides the identification of the MED level), one should apply the GORIC to the set containing all of these.

Supplementary Materials

The supplementary material contains four figures (like Figure 3), each depicting the simulation results for one of the four true dose-response shapes (i.e., linear, convex, semi-concave, and concave). That is, each figure displays the simulated MED selection rates for the three methods (i.e., the GORIC and the Dunnett- and Williams-type multiple contrast test) for one of the true dose-response shapes.

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References

- H. Akaike. Information theory and an extension of the maximum likelihood principle. In B.N. Petrov and F. Csaki, editors, *Second International Symposium on Information Theory*, pages 267–281. Budapest: Akademiai Kiado, 1973.

- K. Anraku. An information criterion for parameters under a simple order restriction. *Biometrika*, 86(1):141–152, 1999.
- G. Aras, A. Xue, and T. Liu. Tukey’s contrast test versus two-sample test in a dose-response clinical trial. *Statistics in Biopharmaceutical Research*, 3(1):31–39, 2011. ISSN 1946-6315.
- P. Bauer. A note on multiple testing procedures in dose finding. *Biometrics*, 53(3):1125–1128, 1997. ISSN 0006-341X.
- F. Bretz. *Powerful Modification of Williams Test on Trend*. PhD thesis, Leibniz University, 1999.
- F. Bretz, T. Hothorn, and P. Westfall. On multiple comparisons in R. *R News*, 2(3):14–17, 2002.
- J. Cohen. A power primer. *Psychological Bulletin*, 112:155–159, 1992.
- Committee for Proprietary Medicine Products. Points to consider on multiplicity issues in clinical trials. Technical report, CPMP/EWP/908/99, 2002.
- G. Dilba, E. Bretz, V. Guiard, and L. A. Hothorn. Simultaneous confidence intervals for ratios with applications to the comparison of several treatments with a control. *Methods of Information in Medicine*, 43(5):465–469, 2004.
- C. W. Dunnett. A multiple comparison procedure for comparing several treatments with a control. *Journal of the American Statistical Association*, 50(272):1096–1121, 1955.

- O. G. Fitzhugh, A. A. Nelson, and M. L. Quaife. Chronic oral toxicity of aldrin and dieldrin in rats and dogs. *Food Cosmetic Toxicology*, 2:551–562, 1964.
- A. Genz and F. Bretz. Numerical computation of multivariate t -probabilities with application to power calculation of multiple contrasts. *Journal of Statistical Computation and Simulation*, 63:361–378, 1999.
- C. Hirotsu, S. Yamamoto, and L. A. Hothorn. Estimating the dose-response pattern by the maximal contrast type test approach. *Statistics in Biopharmaceutical Research*, 3(1):40–53, 2011.
- L. A. Hothorn and M. Hasler. Proof of hazard and proof of safety in toxicological studies using simultaneous confidence intervals for differences and ratios to control. *Journal of Biopharmaceutical Statistics*, 18(5):915–933, 2008. ISSN 1054-3406.
- L. A. Hothorn and D. Hauschke. Identifying the maximum safe dose: A multiple testing approach. *Journal of Biopharmaceutical Statistics*, 10(1):15–30, 2000.
- H. M. J. Hung and S.-J. Wang. Challenges to multiple testing in clinical trials. *Biometrical Journal*, 52:747–756, 2010. ISSN 0323-3847.
- R. L. Kodell. Replace the noael and loael with the bmdl(01) and bmdl(10). *Environmental and Ecological Statistics*, 16(1):3–12, 2009. ISSN 1352-8505.
- R. M. Kuiper. *Model Selection Criteria: How to Evaluate Order Restrictions*. PhD thesis, Utrecht University, 2011.
- R. M. Kuiper. Model selection under inequality constraints in small samples. Unpublished.

- R. M. Kuiper, H. Hoijsink, and M. J. Silvapulle. An Akaike-type information criterion for model selection under inequality constraints. *Biometrika*, 98(2):495–501, 2011.
- R. M. Kuiper, H. Hoijsink, and M. J. Silvapulle. Generalization of the order-restricted information criterion for multivariate normal linear models. *Journal of Statistical Planning and Inference*, 142:2454–2463, 2012.
- W. Leisenring and L. Ryan. Statistical properties of the noel. *Regulatory Toxicology and Pharmacology*, 15(2):161–171, 1992. ISSN 0273-2300.
- S. J. Ruberg. Contrasts for identifying the minimum effective dose. *Journal of the American Statistical Association*, 84(407):816–822, 1989. ISSN 0162-1459.
- S. Sand, C. J. Portier, and D. Krewski. A signal-to-noise crossover dose as the point of departure for health risk assessment. *Environmental Health Perspectives*, 119(12):1766–1774, 2011.
- J. Shang. A multiple comparison procedure based on a variant of the schwarz information criterion in a mixed model. *Communications in Statistics Theory and Methods*, 39:1095–1109, 2010.
- R. W. West and R. L. Kodell. Changepoint alternatives to the noel. *Journal of Agricultural Biological and Environmental Statistics*, 10(2):197–211, 2005. ISSN 1085-7117.
- P. H. Westfall, R. D. Tobias, D. Rom, R. D. Wolfinger, and Y. Hochberg. *Multiple Comparisons and Multiple Tests Using the SAS System*. Cary, NC: SAS Institute Inc., 1999.

D. A. Williams. Test for differences between treatment means when several dose levels are compared with a zero dose control. *Biometrics*, 27(1):103–117, 1971.

T. Yanagawa and Y. Kikuchi. Statistical issues on the determination of the no-observed-adverse-effect levels in toxicology. *Environmetrics*, 12:319–325, 2001. ISSN 1180-4009.

A Calculation of the GORIC

The GORIC is calculated by

$$GORIC = -2 \log L_m + 2 PT_m,$$

with $\log L_m$ the order-restricted log likelihood and PT_m the penalty/complexity term for Hypothesis H_m . The order-restricted likelihood is based on the means that maximize the likelihood subject to the restrictions in Hypothesis/Model H_m . We will refer to these means as the order-restricted means or the order-restricted maximum likelihood estimates (order-restricted MLEs) of $\boldsymbol{\mu}$ and denote them by $\tilde{\boldsymbol{\mu}}$. The order-restricted log likelihood is given by

$$\log L_m = -\frac{1}{2}N \log(2\pi\tilde{\sigma}^2) - \frac{1}{2} \sum_{i=0}^k \sum_{j=1}^{n_i} \left\{ \frac{y_{ij} - \tilde{\mu}_i}{\tilde{\sigma}} \right\}^2,$$

with $N = \sum_{i=0}^k n_i$ and $\tilde{\sigma}$ the order-restricted MLE of σ . The penalty term is calculated by

$$PT_m = 1 + \sum_{l=1}^{k+1} LP(k+1, n_0, \dots, n_k, H_m) \cdot l, \quad (3)$$

where $LP(\cdot)$ is the level probability for Hypothesis H_m , which depends on the number of dose levels ($k+1$, namely 0 to k), the number of observations per dose group (n_i for $i = 0, \dots, k$), and the restrictions in Hypothesis H_m . In case Hypothesis H_m comprises

simple order restrictions, $LP(\cdot)$ is the a priori probability that there are l distinct mean values among the $k + 1$ order-restricted means. Then, the penalty term can be seen as the expected number of distinct mean values plus one (because of the unknown variance term).

The GORIC selects the best out of a set of hypotheses, in which the parameters (here μ_i s) may be restricted by a mixture of linear equality and inequality constraints. It can be applied to multivariate normal linear models with hypotheses of the form

$$H_m : \quad \mathbf{R}_1 \boldsymbol{\mu} \leq \mathbf{r}_1, \mathbf{R}_2 \boldsymbol{\mu} = \mathbf{r}_2, \text{ for } m \in \mathcal{M},$$

where $\mathcal{M}$ is the set of hypothesis indices, $\boldsymbol{\mu}$ a $(k + 1)$ -vector of the $k + 1$ parameters, $\mathbf{R}_1$ a $c_{m1} \times (k + 1)$ matrix containing c_{m1} inequality restrictions on the $k + 1$ parameters, $\mathbf{r}_1$ the right-hand-side of the inequality restrictions of length c_{m1} , $\mathbf{R}_2$ a $c_{m2} \times (k + 1)$ matrix with c_{m2} equality restrictions on the $k + 1$ parameters, and $\mathbf{r}_2$ the right-hand-side of the equality restrictions of length c_{m2} [Kuiper et al., 2011, 2012]. Special cases of these matrix order restrictions are the simple order restrictions, that is,

$$H_m : \quad \mu_0 \leq \dots \leq \mu_k,$$

where “ $\leq$ ” may be replaced with “ $=$ ”, and the tree order, that is, $H_m: \mu_0 \leq \mu_1, \dots, \mu_0 \leq \mu_k$.

For instance, when $k = 2$, the simple order restriction $H_m: \mu_0 \leq \mu_1 \leq \mu_2$ can be written as

$$H_m : \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} \begin{bmatrix} \mu_0 \\ \mu_1 \\ \mu_2 \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

It should be stressed that in case of simple order restrictions, the GORIC reduces to the

ORIC of Anraku [1999]. The only type of restrictions that the GORIC cannot evaluate are range restrictions; for example, $H_m: 2 \leq \mu_1 \leq 3$.

One can best construct the set of hypotheses based on theory. This (often) leads to a limited set of (reasonable) hypotheses. One should include the unconstrained model to safeguard from choosing a weak hypothesis as the best one. In the identification of a MED or LOAEL, one could examine the set where one hypothesis contains all possible orderings representing a specific dose to be the MED. Alternatively, one can evaluate the hypotheses resulting from the decomposition of the global order-restricted alternative in case one is interested in the pattern of the response means as well.

B GORIC weights

Since the likelihood increases with the number of observations, the GORIC values themselves are not of interest. To improve interpretation, we also inspect the GORIC weights (w_m). Another reason to use the weights might be that Shang [2010] shows that the use of probabilities (like the GORIC weights) is a more effective procedure for ‘multiple comparisons’ than that based on the information criterion itself.

The weight w_m is calculated by:

$$w_m = \frac{\exp(-1/2 (GORIC_m - GORIC_{min}))}{\sum_{m' \in \mathcal{M}} \exp(-1/2 (GORIC_{m'} - GORIC_{min}))},$$

where $\mathcal{M}$ is the set of (say, M) hypothesis indices and $GORIC_{min}$ is the lowest GORIC value, that is, the GORIC value of the preferred model/hypothesis.

Because the GORIC can be seen as a likelihood for Hypothesis H_m , the GORIC weight represents the relative likelihood (or, stated otherwise, the weight of evidence) of Hypoth-

esis H_m given the data and the set of (say, M) hypotheses. Hence, the weights reflect the relative support of one hypothesis of interest in comparison to the whole set. Moreover, the ratio of two weights gives the relative support of these two hypotheses, leading to statements as ‘ H_m is $w_m/w_{m'}$ fold more likely than $H_{m'}$ ’.